Supplementary material of the article titled: Comparison of Spectral Multi-material Phase Retrieval Algorithms based on Edge Illumination Single Mask X-ray Phase Contrast Imaging

1. Experimental comparison

To evaluate the suitability of both approaches while accounting for different experimental non-idealities, an experimental study was carried out. An additional simulation replicating the experimental conditions previously presented was performed. The experimental and simulation setups are illustrated schematically in Figure S1.

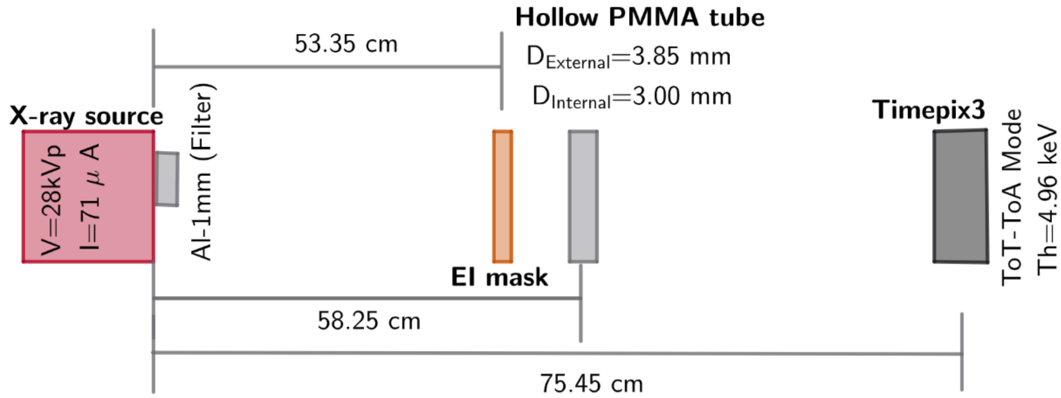


Figure S1. Experimental and simulation setups, where the X-ray source, the sample, the mask, and the detector are illustrated.

In the experimental study, a micro-focus Hamamatsu L6622-01 was used, operated at a current of 71 μA and a voltage of 28 kVp. The corresponding focal spot size was 13.81 μm, according to previous experimental measurements [S5]. A 1 mm thick Aluminum filter was employed to suppress the contribution of photons with energies lower than 10 keV.

The X-ray detector employed was a Timepix3, positioned 75.45 cm from the X-ray source. It is paired to a 1 mm thick CdTe sensor, which consists of a 256x256 matrix of square pixels with a pixel pith of 55 μm [S4]. The detector was operated with a bias voltage of -300 V and an energy threshold of 4.96 keV.

The sample consisted of a hollow PMMA tube, with internal (external) diameter of 3 mm (3.85 mm). The sample was positioned downstream of the EI mask, at a distance of 58.25 cm from the X-ray source. The EI mask is composed of a periodic structure consisting of micrometer gold rods, galvanized on a graphite substrate 500 μm thick and regularly spaced with a period of 79 μm [S4]. To achieve the EI condition, the mask was positioned at 53.35 cm from the X-ray source. Eight dithering steps were required to obtain the complete image.

Sample images were acquired using the ToT (time-over-threshold)-ToA (time of arrival) mode of the detector from a single X-ray exposure. To correct the charge sharing effect, a clustering algorithm was employed [S1]. The exposure time per dithering step was 15 minutes, resulting in a total exposure time of 120 minutes for the final sample image. To construct the $F_{\mp}$ matrix of equation 8, each sample image was divided into 4 keV energy bins, yielding $\vartheta = 4$ energy-bin sample images $I_{z_1}(\vec{r}_e, \omega_\vartheta)$ corresponding to the ranges 13–16 keV, 17–20 keV, 21–24 keV, and 25–28 keV. The average energy for each bin was computed from the detected photon energy distribution within the corresponding range, yielding mean energies of 15, 18, 22, and 26 keV, respectively. The energy-bin sample images $I_{z_1}(\vec{r}_e, \omega_\vartheta)$ were split into even and odd column images to obtain $I_{z_1-}(\vec{r}_e, \omega_\vartheta)$ and $I_{z_1+}(\vec{r}_e, \omega_\vartheta)$, respectively.

Reference images were required to complete the construction of the $F_\mp$ matrix. These were obtained by acquiring images without the sample using the same exposure time as for the sample images. The reference acquisition was then divided into energy bins, yielding $\vartheta = 4$ energy-bin reference images $I_{oz_1}(\vec{r}_e, \omega_\vartheta)$ and leading to $I_{-Z_o}\left(\frac{\vec{r}_e}{M}, \omega_\vartheta\right)$ through the relation $I_{-Z_o}\left(\frac{\vec{r}_e}{M}, \omega_\vartheta\right) = M^2 I_{oz_1}(\vec{r}_e, \omega_\vartheta)$. Similarly, the energy-bin reference images $I_{oz_1}(\vec{r}_e, \omega_\vartheta)$ were split into even and odd column images to obtain $I_{oz_1-}(\vec{r}_e, \omega_\vartheta)$ and $I_{oz_1+}(\vec{r}_e, \omega_\vartheta)$, respectively. The energy-bin flat-field (FF)-corrected images were defined as $CI_\mp = \frac{I_{z_1\mp}(\vec{r}_e, \omega_\vartheta)}{I_{oz_1\mp}(\vec{r}_e, \omega_\vartheta)}$. Dead pixels (around 5\% of the entire detector, located mainly at the edges and the upper right region of the sensor) were masked in the images, and the average pixel intensity of their nearest neighbors was used instead.

To validate the experimental results, similar conditions were used in the simulation study. Specifically, the detector was based on the Timepix3 detector, modeled as a 256x256 pixel array with a pixel pith of 55 μm and CdTe as the sensor material. It was positioned at 75.45 cm from the X-ray source, and the calibrated energy resolution used in the corresponding code was based on the calibrated resolution of the Timepix3. Additionally, the X-ray source was modeled as a circular source with a FWHM of 13.81 μm. This source emits primary X-ray photons in a conical beam with an opening angle θ and a radius R at the detector position. To ensure that the detector area was fully circumscribed within the circular cross-sectional area of the beam cone, θ was set to 13 mrad. The X-ray source was monochromatic, taking energy-bin images at 15, 18, 22 and 26 keV. The number of photons for each energy bin was determined by the weight of a spectrum, such as $N_0(E) = w(E)N_o$, where N_o is the total number of emitted photons and $w(E)$ is the weight of the spectrum at E. Similar to the experimental study, a normalized spectrum from a tungsten anode with a 1 mm thick aluminum filter at 28 kVp was used, obtained from the TASMICS spreadsheet [S2]. Furthermore, the total number of emitted photons from the X-ray source used to acquire both the energy-bin sample images and the energy-bin reference images was $N_o = 1.1 \times 10^{10}$, and the corresponding energy-bin FF-corrected images were obtained by following the same procedure as in the experimental study.

The sample emulated the hollow PMMA tube using two concentric cylinders; the inner cylinder of Air (diameter of 3.00 mm) and the outer cylinder of PMMA (diameter of 3.85 mm). The materials in Geant4 were defined by specifying their densities in $\frac{g}{cm^3}$ (PMMA=1.19 and Air=0.0012), and their elemental weight percentages (PMMA: H=8.05, C=59.99, O=31.96; Air: C=0.01, N=75.53, O=23.18, Ar=1.28) [S1]. Furthermore, the M_1 mask in the simulation study was modeled as a periodic structure of micrometer gold bars, with the same period, bar width, and aperture length as the experimental M_1 mask.

The application of equations 9, 11, 12, and 15 allowed for the retrieval of the projected thickness using the different approaches. The same process presented in the manuscript was followed in both the experimental and simulations studies. The prior knowledge of the total projected thickness was derived from the dimension of the external diameter; therefore, a solid cylinder with a diameter of 3.85 mm was built in Python and used as Z_o in equations 6 and 13. As a single material is embedded (Air) within the encasing material (PMMA), the information of the $\vartheta = 4$ input images was used in equations 9, 11, 12, and 15 to retrieve the projected thickness of Air, with $\Omega - 1 = 1$ and $\Delta\beta_2 = \beta_{Air} - \beta_{PMMA}$ ($\Delta\delta_2 = \delta_{Air} - \delta_{PMMA}$). To construct the Υ matrix and the ζ function, $\bar{S}_B$ and σ_B were calculated in red circular region of interest (ROI), while $\bar{S}_R$ was calculated in the blue ROI, as illustrated in the following CI_{-} at the energy-bin 13–16 keV (Figure S2).

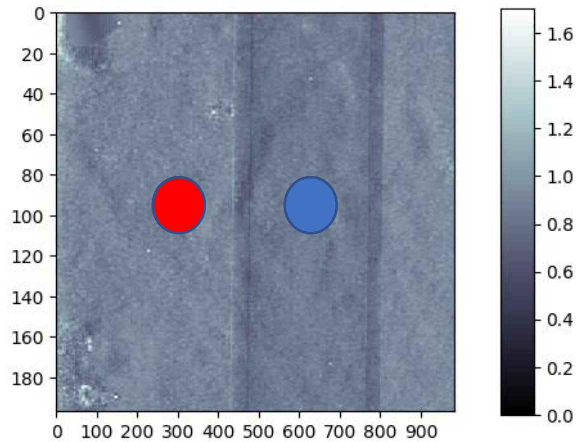
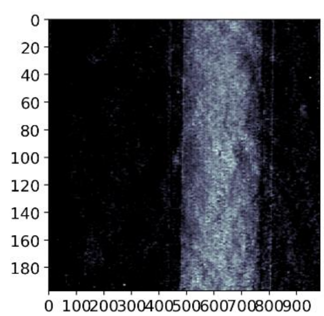
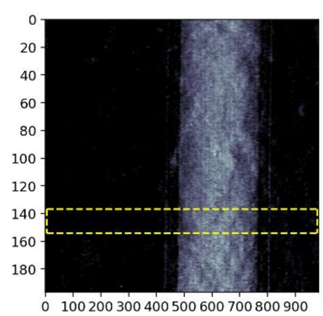
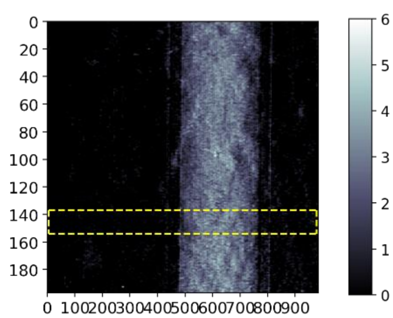


Figure S2. Visualization of the regions selected in the CI_{-} image at the energy-bin 13–16 keV for evaluating the Υ matrix and the ζ function.

Figure S3 presents the projected thickness retrieved by the Beltran approach at the energy-bin 17–20 keV (left), the spectral multi-material decomposition approach (center) and the energy-weighted Beltran approach (right). The first column illustrates experimental results, while the second column shows simulation results.

Beltran approach (17-20 keV)	Spectral approach	Energy-weighted Beltran approach
		

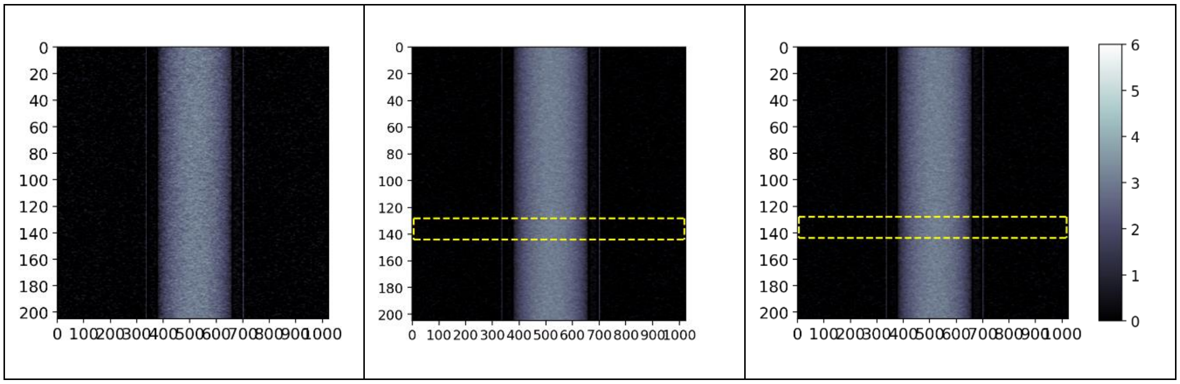


Figure S3. Retrieved projected thickness map of air obtained using the Beltran approach at the energy-bin 17-20 keV (left), the spectral multi-material decomposition approach (center) and the energy-weighted Beltran approach (right) with the $CI_{\pm}$ images. The first column corresponds to the results obtained in the experimental study, whereas the second column corresponds to the results obtained in the simulation part.

To evaluate the trend of the projected thickness retrieved for each approach, the projected thickness profiles are depicted in Figure S4, defined as the projections of the thickness maps within the selected yellow regions. The expected thickness profile of a cylinder with a diameter of 3.00 mm is included for comparison (green dotted line).

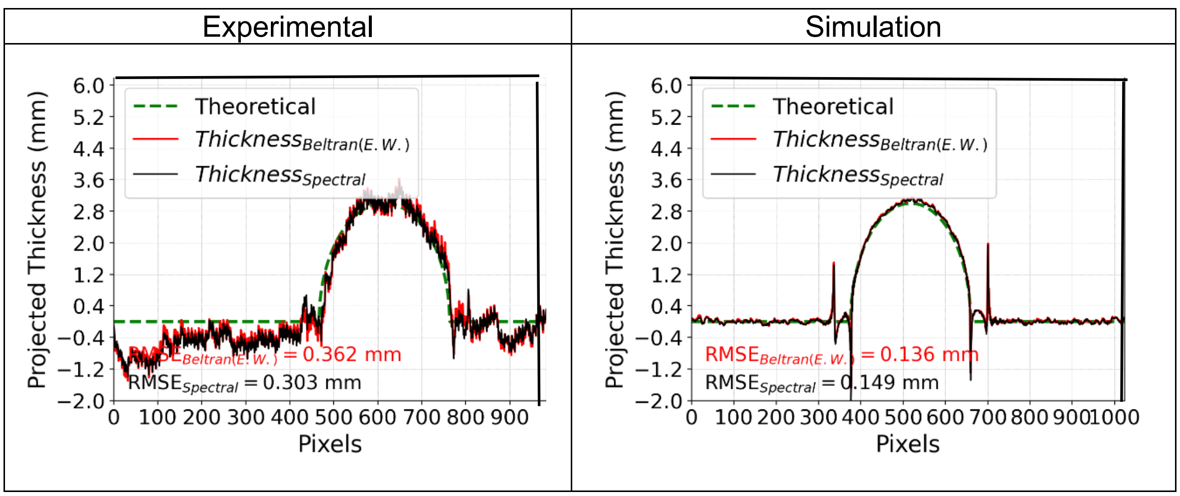


Figure S4. Comparison of the retrieved and theoretical projected thickness profiles. The left part corresponds to the results obtained in the experimental study, whereas the right part corresponds to the results obtained in the simulation part.

The agreement between the retrieved profiles and the theoretical profile in both experimental and simulation studies indicates that all approaches closely follow the expected behavior, as further supported by the small RMSE (root mean square error) values reported in the previous figure. Higher RMSE values are observed in the experimental profiles due to the higher noise levels in the experimental images. Furthermore, the minor discrepancies observed in the experimental trends outside the sample (i.e., negative retrieved thickness) are attributable to the use of energy-bin images with a 4 keV bin width. Although the number of energies contributing to each energy-bin image is small, the combined contributions lead to an absorption variation that differs from that predicted using the assumed mean energy.

Analyzing the CNR values in the projected thickness maps, calculated using the same regions for the ζ function, Figure S5 illustrates that the CNR values from simulations are higher than those from experiments. This difference arises from the idealized nature of the simulated data, in which only Poisson noise is typically considered. Experimental non-idealities introduce additional noise sources that increase the background standard deviation and thereby reduce the CNR in the experimental results. However, the results presented in the manuscript are supported by both experimental and simulation studies. Specifically, the CNR of the thickness maps is higher than those in the CI images, and the CNR obtained using the spectral multi-material decomposition approach is higher than that obtained with the Beltran and energy-weighted Beltran approaches. This demonstrates that the spectral approach outperforms the Beltran approach in visualizing features with higher CNR when the same number of $CI_{\pm}$ images and equivalent absorbed dose conditions are used to estimate the projected thickness of a single material within the sample.

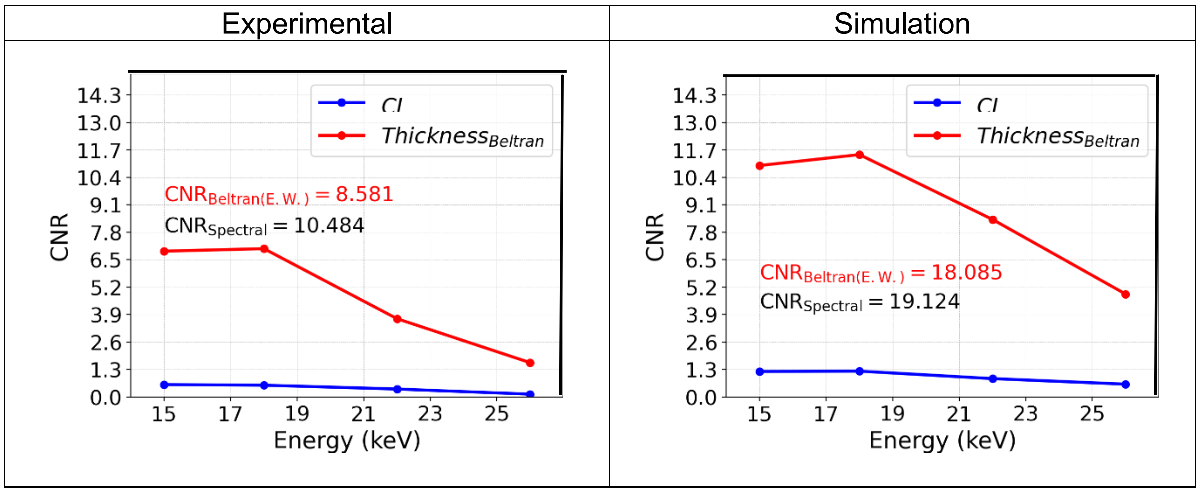


Figure S5. CNR of the inner cylinder in the $CI_{\pm}$ images (blue curve) and in the projected thickness maps obtained using the Beltran approach (red curve) at each energy-bin. CNR values obtained by the spectral multi-material decomposition and the energy-weighted Beltran approaches are also illustrated. The left part corresponds to the results obtained in the experimental study, whereas the right part corresponds to the results obtained in the simulation part.

2. Simulation-Materials

Material	Density $\left(\frac{g}{cm^3}\right)$	Elemental weight
Bone	1.45	$H \approx 6.55\%$, $C \approx 53.69\%$, $N = 2.15\%$, $O \approx 3.21\%$, $F \approx 16.74\%$, $Ca \approx 17.66\%$
Air	0.0012	$C \approx 0.012\%$, $N \approx 75.53\%$, $O \approx 23.18\%$, $Ar \approx 1.28\%$
Muscle	1.05	$H = 10.20\%$, $C = 14.30\%$, $N = 3.40\%$,

		$O = 71.00\%$, $Na = 0.10\%$, $P = 0.20\%$ $S = 0.30\%$ $Cl = 0.10\%$ $K = 0.40\%$
Skin	1.09	$H \approx 10.01\%$, $C \approx 20.42\%$, $N \approx 4.19\%$, $O \approx 64.57\%$, $Na \approx 0.18\%$, $P \approx 0.08\%$, $S \approx 0.16\%$ $Cl \approx 0.27\%$ $K \approx 0.10\%$
Fat layer (Adipose)	0.95	$H \approx 11.39\%$, $C \approx 59.78\%$, $N \approx 0.71\%$, $O \approx 27.79\%$, $Na \approx 0.13\%$, $S \approx 0.09\%$ $Cl = 0.10\%$

Table S_1. Information of materials used in PEPI code, providing their density and their elemental weight [S6,S7].

References

[S1] E. Cely, G. Roque, S. Useche, C. Ávila and S. Procz, “CNR enhancement in breast implant phantoms via spectral x-ray imaging,” Phys. Med. Biol., vol. 71, no. 1, 015013, 2025. [Online]. Available: https://iopscience.iop.org/article/10.1088/1361-6560/ae29df

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